

Andres Perez

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Aspiring Embedded Systems Engineer interested in Front-End ASIC design and FPGA prototyping

EDUCATION

University of Notre Dame, Notre Dame, IN

Graduation: May 2028

Bachelor of Science in Computer Engineering & Mathematics minor | QuestBridge Scholar | Fall 2025 Dean's List

GPA: 3.76

Engineering Study Abroad: London, England

May 2025 – June 2025

- **Relevant Coursework:** Data Structures, **System Programming, Logic and Processor Design, Signals and Information Systems,** Digital Design for Smart Interconnected Systems, Electronics Devices & Systems, Intro to Artificial Intelligence

ENGINEERING LEADERSHIP

IrishSat's Gravitational Orbital Attitude Thermal Lab (GOAT Lab)

Notre Dame, IN

Electronics Lead | CubeSat Team

Aug. 2024 – Present

- Spearheading a team of 6+ engineers for the embedded systems development and PCB design initiatives for testing equipment, particularly a Helmholtz cage project, coordinating hardware-software integration across CubeSat projects.
- Collaborated under expert mentorship to refine IrishSat's Helmholtz Cage design, integrating PCB design (KiCad), control algorithms (MATLAB), and embedded Linux systems (Raspberry Pi) into a unified testing platform.
- Designed and assembled a custom 7.2V 7000 mAh 2S2P Li-ion battery pack using 18650 cells and spot welding.

ENGINEERING PROJECTS

Hardware-Accelerated Human Activity Classifier | Independent Project

Long Beach, CA

Designer

July 2025 – Aug. 2025

- Built a wireless data acquisition system using two Arduino Uno R3s, an MPU-9250 IMU sensor, and nRF24L01 RF transceivers.
- Developed a C++ program to stream accelerometer/gyroscope data wirelessly and a Python script to parse it into a labeled walking/sitting dataset.
- Achieved 96.7% classification accuracy with a PyTorch 1D CNN, documenting performance across various hyperparameters.

Helmholtz Cage Driver PCB | GOAT Lab

Notre Dame, IN

Lead Designer

Mar. 2025

- Designed a custom Arduino R3 hat PCB using KiCad to integrate and optimize Arduino, magnetometers, and H-bridge circuits, reducing assembly complexity.
- Amended the generated Bill of Materials and Component Placement List files export to JLCPCB for manufacturing.

Portfolio Website | andres-perez.github.io

Long Beach, CA

Developer

Aug. 2024 - Present

- Developed a portfolio website with Astro and Tailwind CSS frameworks as the front end to showcase engineering achievements.

RESEARCH & WORK EXPERIENCE

Engineering Career Assistant | Meruelo Family Center for Career Development

Sept. 2025 – Present

- Coauthoring a newsletter for undergraduates, specifically seniors, highlighting career opportunities and post-graduate experiences from engineering alumni.
- Grew the newsletter to a best 81.70% open rate with a 3.89% click rate of 517 recipients.

Student Researcher | Notre Dame Department of Electrical Engineering

Aug. 2025 – Nov. 2025

- Designed a real-time client-server interactive map to parse GPS data and track tether ORION drones.
- Analyzed drone telemetry traffic using Wireshark to validate GPS data was transmitted via MAVLink rather than NMEA strings; Integrated PyMAVLink to extract and forward real-time telemetry from drones to the visualization client.

SKILLS & LANGUAGE

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|---|------------------------------|--------------------|---|-------------------------|-------------|
| ▪ C / C++ | ▪ Python | ▪ SystemVerilog | ▪ MATLAB | ▪ Assembly | ▪ Shell |
| ▪ Git / GitHub | ▪ Vim | ▪ Circuit Analysis | ▪ KiCad / Altium | ▪ SolidWorks | ▪ Soldering |
| ▪ Technical Writing | ▪ Wireshark Network Analyzer | | ▪ Microsoft Office Suite (Excel, PowerPoint, Word, Teams) | | |
| ▪ Electronics Test Bench (Function Generator, Oscilloscope, Multimeter, LCR meter, Power Supplies, Electronics Soldering) | | | | | |
| ▪ English (Native) | | ▪ Spanish (Native) | | ▪ Japanese (Elementary) | |